Week 10 Homework

October 18, 2026

Question 1

(a)

we are given:

- metal sheet of thickness d with mobile electron density n
- electrons displaced by $\Delta \ll d$ along the x axis

displacing the electrons leaves a sheet of positive charge on one side and a sheet of negative charge on the other. each sheet has a thickness Δ, so the charge per unit area is

$$\sigma = ne\Delta$$

the hint tells us to consider a small cylinder that encloses a piece of the positive sheet. by symmetry, the field points away from the sheet on both sides so the flux through the two ends is $2EA_c$ where A_c is the end surface area of the cylinder. using Gauss' law,

$$\oint E \cdot dA = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

$$2E_+A_c = \frac{\sigma A_c}{\varepsilon_0}$$

$$E_+ = \frac{\sigma}{2\varepsilon_0}$$

the negative sheet gives the same magnitude, so,

$$E_- = \sigma/2\varepsilon_0$$

between the two sheets, both fields point from the positive sheet towards the negative sheet, so they add

$$E = E_+ + E_- = \frac{\sigma}{\varepsilon_0}$$

substituting $\sigma = ne\Delta$,

$$\boxed{E = \frac{ne\Delta}{\varepsilon_0}}$$

(b)

once the electrons are released, each electron feels a force from the field. the field points from the positive sheet to the negative sheet, so the force on an electron points back towards equilibrium

$$F = -eE = -\frac{ne^2\Delta}{\varepsilon_0}$$

using $F = m_e\ddot{\Delta}$,

$$m_e\ddot{\Delta} = -\frac{ne^2}{\varepsilon_0}\Delta$$

$$\boxed{\ddot{\Delta} + \frac{ne^2}{m_e\varepsilon_0}\Delta = 0}$$

this is the equation for undamped SHM.

(c)

comparing with $\ddot{x} + \omega_0^2 x = 0$,

$$\omega_0^2 = \frac{ne^2}{m_e\varepsilon_0}$$

$$\boxed{\omega_0 = \sqrt{\frac{ne^2}{m_e\varepsilon_0}}}$$

(d)

we are given:

- density: $\rho = 10.5\,\mathrm{g\,cm^{-3}}$
- atomic mass: $M = 108\,\mathrm{g\,mol^{-1}}$
- one mobile electron per atom

since there is one mobile electron per atom, n is the number density of atoms,

$$n = \frac{\rho N_A}{M}$$

$$n = \frac{10.5}{108}(6.022 \times 10^{23})$$

$$n \approx 5.85 \times 10^{22}\,\mathrm{cm}^{-3} = 5.85 \times 10^{28}\,\mathrm{m}^{-3}$$

substituting into the result from (c),

$$\omega_0^2 = \frac{(5.85 \times 10^{28})(1.602 \times 10^{-19})^2}{(9.109 \times 10^{-31})(8.854 \times 10^{-12})}$$

$$\omega_0^2 \approx 1.86 \times 10^{32}\,\mathrm{s}^{-2}$$

$$\boxed{\omega_0 \approx 1.36 \times 10^{16}\,\mathrm{rad\,s}^{-1}}$$

the frequency is

$$f = \frac{\omega_0}{2\pi} \approx 2.17 \times 10^{15}\,\mathrm{Hz}$$

the wavelength of light with this frequency is

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega_0}$$

$$\lambda = \frac{2\pi(2.998 \times 10^8)}{1.365 \times 10^{16}}$$

$$\lambda \approx 1.38 \times 10^{-7}\,\mathrm{m}$$

$$\boxed{\lambda \approx 138\,\mathrm{nm}}$$

this is in the ultraviolet.

Question 2

(a)

we are given:

- bridge height: $H = 100\,\mathrm{m}$
- the cord stretches by 20% when the jumper hangs from it
- the fall stops exactly at the river

let the unstretched length of the cord be L.

when the jumper hangs at rest, the spring force balances gravity and the extension is $0.2L$,

$$k(0.2L) = mg$$
$$k = \frac{5mg}{L}$$

at the lowest point the jumper is at the river, so the cord has been stretched by

$$x_{\max} = H - L$$

the jumper is at rest at the top and at the bottom, so all the gravitational potential energy lost goes into the elastic energy of the cord,

$$mgH = \frac{1}{2}k(H - L)^2$$

substituting $k = 5mg/L$,

$$mgH = \frac{5mg}{2L}(H - L)^2$$
$$100 = \frac{5}{2L}(100 - L)^2$$
$$40L = (100 - L)^2$$
$$L^2 - 240L + 10000 = 0$$

using the quadratic formula,

$$L = \frac{240 \pm \sqrt{240^2 - 4(10000)}}{2}$$
$$L = \frac{240 \pm \sqrt{17600}}{2}$$
$$L \approx 53.7\,\text{m} \quad \text{or} \quad L \approx 186\,\text{m}$$

the cord cannot be longer than the bridge height, so we reject the second root,

$$\boxed{L \approx 53.7\,\text{m}}$$

(b)

after the oscillations stop the jumper hangs at rest, where the cord is stretched by 20%,

$$\text{extension} = 0.2L \approx 10.7\,\text{m}$$

so the total length of the hanging cord is

$$1.2L \approx 64.4\,\text{m}$$

the height above the river is

$$h = H - 1.2L$$

$$h = 100 - 64.4$$

$$\boxed{h \approx 35.6\,\text{m}}$$

(c)

for small oscillations about the equilibrium position the cord stays taut, so the jumper behaves as a mass on a spring,

$$\omega_0 = \sqrt{\frac{k}{m}}$$

the period is

$$T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}$$

using $k = 5mg/L$,

$$T = 2\pi\sqrt{\frac{L}{5g}}$$

$$T = 2\pi\sqrt{\frac{53.67}{5(9.81)}}$$

$$T \approx 6.57\,\text{s}$$

$$\boxed{T \approx 6.6\,\text{s}}$$

Question 3

(a)

we are given:

- resonant frequency: $f_0 = 19.7\,\text{MHz}$
- mass: $m = 9.7 \times 10^{-16}\,\text{kg}$
- quality factor: $Q = 40000$

the angular frequency is

$$\omega_0 = 2\pi f_0$$
$$\omega_0 = 2\pi(19.7 \times 10^6)$$
$$\omega_0 \approx 1.238 \times 10^8\,\text{rad}\,\text{s}^{-1}$$

since $\omega_0^2 = k/m$,

$$k = m\omega_0^2$$
$$k = (9.7 \times 10^{-16})(1.238 \times 10^8)^2$$
$$\boxed{k \approx 14.9\,\text{N}\,\text{m}^{-1}}$$

(b)

the quality factor is

$$Q = \frac{\omega_0}{\gamma}$$

so,

$$\gamma = \frac{\omega_0}{Q}$$
$$\gamma = \frac{1.238 \times 10^8}{40000}$$
$$\boxed{\gamma \approx 3.09 \times 10^3\,\text{s}^{-1}}$$

(c)

since Q is large, we have $\omega_0/\gamma \gg 10$, so we can use $\omega' \approx \omega_0$. the undriven motion is a decaying oscillation,

$$x(t) \approx A_0 e^{-\gamma t/2} \cos(\omega_0 t)$$

the energy of the oscillator is

$$E = \frac{1}{2} k A^2$$

and the amplitude decays as $A = A_0 e^{-\gamma t/2}$, so

$$E(t) = \frac{1}{2} k A_0^2 e^{-\gamma t}$$

differentiating,

$$\frac{dE}{dt} = -\gamma E$$

so,

$$\boxed{\frac{dE}{dt} = -\frac{\gamma}{2} k A_0^2 e^{-\gamma t}}$$

at $t = 0$ the energy is lost at a rate $\gamma E_0 = \frac{1}{2}\gamma k A_0^2 = \frac{1}{2} m \gamma \omega_0^2 A_0^2$.

(d)

on resonance $\omega = \omega_0$, so the steady state amplitude is

$$A = \frac{F_0}{m\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$$

$$A = \frac{F_0}{m \gamma \omega_0}$$

rearranging for the driving force,

$$F_0 = m \gamma \omega_0 A$$

substituting, with $A = 400\,\text{fm} = 4 \times 10^{-13}\,\text{m}$,

$$F_0 = (9.7 \times 10^{-16})(3.09 \times 10^3)(1.238 \times 10^8)(4 \times 10^{-13})$$

$$\boxed{F_0 \approx 1.49 \times 10^{-16}\,\text{N}}$$

this is the same as kA/Q, as expected.

(e)

the drive frequency is

$$\omega = 2\pi(19.7001 \times 10^6)$$

so the drive is above resonance, with

$$\omega - \omega_0 = 2\pi(100) \approx 628\,\mathrm{rad\,s^{-1}}$$

the phase of the position relative to the drive is

$$\phi = -\arg\left(\omega_0^2 - \omega^2 + i\gamma\omega\right)$$

since $\omega \approx \omega_0$,

$$\omega_0^2 - \omega^2 \approx -2\omega_0(\omega - \omega_0)$$
$$\omega_0^2 - \omega^2 \approx -2(1.238 \times 10^8)(628)$$
$$\omega_0^2 - \omega^2 \approx -1.555 \times 10^{11}\,\mathrm{s^{-2}}$$

and,

$$\gamma\omega \approx (3.09 \times 10^3)(1.238 \times 10^8)$$
$$\gamma\omega \approx 3.83 \times 10^{11}\,\mathrm{s^{-2}}$$

the real part is negative and the imaginary part is positive, so the complex number lies in the second quadrant,

$$\arg\left(\omega_0^2 - \omega^2 + i\gamma\omega\right) = \pi - \arctan\left(\frac{3.83}{1.555}\right)$$
$$\arg\left(\omega_0^2 - \omega^2 + i\gamma\omega\right) \approx 1.957\,\mathrm{rad}$$

hence,

$$\boxed{\phi \approx -1.96\,\mathrm{rad} \approx -112^\circ}$$

so the position lags the drive by about 112°.

the velocity is $\dot{x} = i\omega x$, so there is an extra phase of $\pi/2$ compared to the position,

$$\theta = \phi + \frac{\pi}{2}$$
$$\theta \approx -1.957 + 1.571$$

$$\boxed{\theta \approx -0.39\,\mathrm{rad} \approx -22^\circ}$$

so the velocity lags the drive by about 22°.

this agrees with the angles for $\omega > \omega_0$, where $-\pi < \phi < -\pi/2$ and $-\pi/2 < \theta < 0$.

(f)

with honey, $\gamma = 4\omega_0$, so

$$\frac{\gamma}{2} = 2\omega_0 > \omega_0$$

this is the over-damped case. the general solution is

$$x(t) = C_1 e^{\alpha_1 t} + C_2 e^{\alpha_2 t}$$

where

$$\alpha_{1,2} = -\frac{\gamma}{2} \pm \sqrt{\left(\frac{\gamma}{2}\right)^2 - \omega_0^2}$$

substituting $\gamma = 4\omega_0$,

$$\alpha_{1,2} = -2\omega_0 \pm \sqrt{4\omega_0^2 - \omega_0^2}$$

$$\alpha_{1,2} = -2\omega_0 \pm \sqrt{3}\,\omega_0$$

so,

$$\alpha_1 = -(2 - \sqrt{3})\omega_0 \approx -0.268\omega_0$$

$$\alpha_2 = -(2 + \sqrt{3})\omega_0 \approx -3.73\omega_0$$

now we use the initial conditions, $x(0) = 0$ and $\dot{x}(0) = v_0$.

$$x(0) = C_1 + C_2 = 0$$

$$C_2 = -C_1$$

$$\dot{x}(0) = \alpha_1 C_1 + \alpha_2 C_2 = v_0$$

$$C_1(\alpha_1 - \alpha_2) = v_0$$

since $\alpha_1 - \alpha_2 = 2\sqrt{3}\,\omega_0$,

$$C_1 = \frac{v_0}{2\sqrt{3}\,\omega_0}$$

therefore,

$$\boxed{x(t) = \frac{v_0}{2\sqrt{3}\,\omega_0}\left(e^{-(2-\sqrt{3})\omega_0 t} - e^{-(2+\sqrt{3})\omega_0 t}\right)}$$

(g)

for $t \gg 1/\omega_0$ the term with α_2 has already died away, since $|\alpha_2| > |\alpha_1|$. so only the slower term remains,

$$x(t) \approx \frac{v_0}{2\sqrt{3}\,\omega_0} e^{-(2-\sqrt{3})\omega_0 t}$$

the decay rate of the amplitude is therefore

$$\boxed{(2-\sqrt{3})\omega_0 \approx 0.268\omega_0 \approx 3.3 \times 10^7\,\mathrm{s}^{-1}}$$

the energy, which goes as x^2, decays at twice this rate.

for comparison, without honey the amplitude decays at $\gamma/2 \approx 1.5 \times 10^3\,\mathrm{s}^{-1}$, so the honey makes the decay about 2×10^4 times faster.

(h)

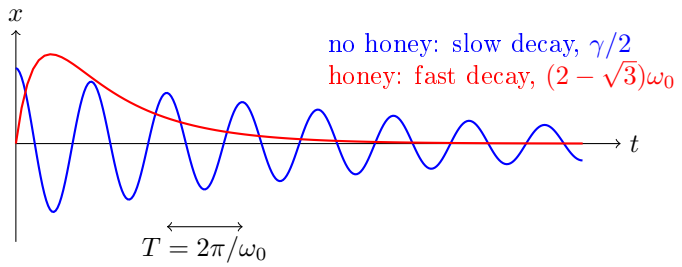
without honey, the resonator is under-damped, so the motion is an oscillation with period

$$T = \frac{2\pi}{\omega_0}$$

inside an envelope that decays slowly at rate $\gamma/2$.

with honey, the resonator is over-damped, so there is no oscillation. the position goes to a single maximum and then decays at rate $(2-\sqrt{3})\omega_0$, which is much faster than the decay without honey.

(just to mention, ai was NOT used to make this graph, i was having some fun :3)



the sketch is not to scale in time for the no honey envelope, since the real decay rates differ by a factor of about 2×10^4. the honey curve is drawn using the true ratio between ω_0 and its decay rate.